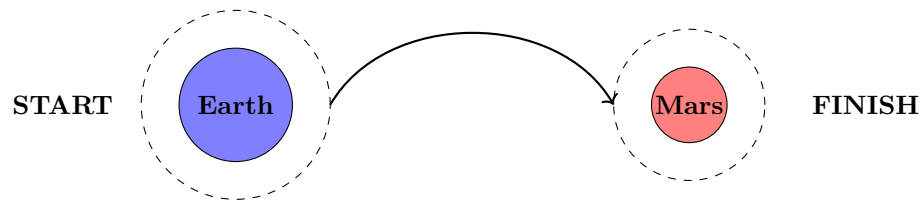


Mission to Mars

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March 1, 2025

Goal: Reach Mars. We will identify the optimal maneuver by evaluating various metrics, including weight, cost, and mission duration. We will end with very complicated multivariable function and try to optimise it.



Boundary Conditions:

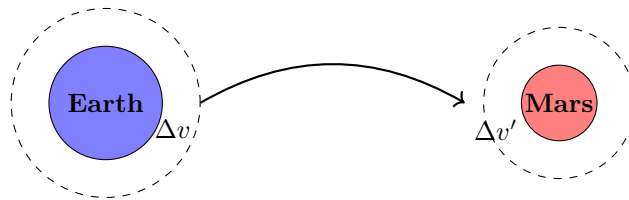
- Earth's Orbit
- Mars' Orbit

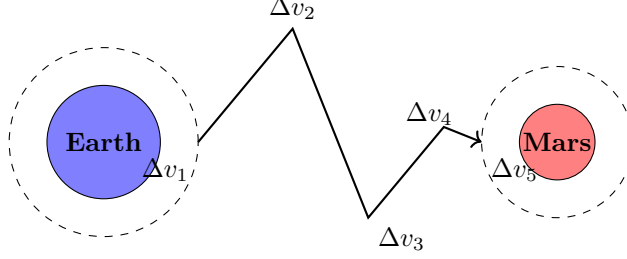
1 Orbital Maneuvers

We are interested in the following types of trajectories:

- Hohmann Transfer
- Impulsive Trajectories (piecewise continuous)
- Continuous Small Energy Addition

In short: many small impulses vs. Hohmann — two large impulses.





In this work, we focus on the equations of n-body dynamics—more precisely, the 3-body problem: Earth, Mars, and a satellite. One of them (the satellite) has negligible mass compared to the other bodies, so the model simplifies to a “constrained” 3-body problem.

The n-body problem considers point masses m_i for $i = 1, 2, \dots, n$ in an inertial reference frame. For our purposes, this will be a heliocentric system where the X-axis is directed toward the constellation Aries and the Y-axis toward the winter solstice (Perihelion).

Summing over all masses, we obtain the equations of motion for the n-body system:

$$m_i \frac{d^2 \mathbf{q}_i}{dt^2} = \sum_{j=1, j \neq i}^n \frac{G m_i m_j (\mathbf{q}_j - \mathbf{q}_i)}{\|\mathbf{q}_j - \mathbf{q}_i\|^3} = -\frac{\partial U}{\partial \mathbf{q}_i} \quad (1)$$

where U is the potential energy of the system.

2 Dynamical Systems

In general, we consider an autonomous system of differential equations of the form:

$$\dot{y} = f(y)$$

Hamiltonian systems, which conserve the system’s energy from a physical point of view, play a crucial role. A Hamiltonian system is a dynamical system that describes the evolution of a physical system (e.g., a planetary system or an electromagnetic field). Such systems are defined as follows:

$$\dot{p} = -H_q(p, q), \quad \dot{q} = H_p(p, q), \quad (2)$$

where $H(p_1, \dots, p_d, q_1, \dots, q_d)$ represents the total energy; q_i are the position coordinates and p_i are the momenta for $i = 1, \dots, d$ with d degrees of freedom. We say that the Hamiltonian H is a **first integral** of such a system.

3 Symplectic Transformations

One of the properties of Hamiltonian systems, as mentioned earlier, is that the Hamiltonian $H(p, q)$ is a first integral of the system.

In this section, we examine another important property—the symplecticity of its phase flow. The objects under consideration are two-dimensional parallelograms lying in $\mathbb{R}^{2d}$.

A parallelogram P is spanned by two vectors:

$$\xi = \begin{pmatrix} \xi^p \\ \xi^q \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta^p \\ \eta^q \end{pmatrix}$$

in the (p, q) space, where $(\xi^p, \xi^q, \eta^p, \eta^q) \in \mathbb{R}^{2d}$.

For $d = 1$, we consider the oriented area of the parallelogram:

$$\text{area}(P) = \det \begin{pmatrix} \xi^p & \eta^p \\ \xi^q & \eta^q \end{pmatrix} = \xi^p \eta^q - \xi^q \eta^p$$

In higher dimensions, we replace the area with the sum of oriented areas of the projections of P onto the coordinate planes (p_i, q_i) :

$$\omega(\xi, \eta) = \sum_{i=1}^d \det \begin{pmatrix} \xi_i^p & \eta_i^p \\ \xi_i^q & \eta_i^q \end{pmatrix} = \sum_{i=1}^d (\xi_i^p \eta_i^q - \xi_i^q \eta_i^p)$$

This defines the bilinear form $\omega(\xi, \eta)$ on vectors from $\mathbb{R}^{2d}$, which plays a central role in Hamiltonian systems:

$$\omega(\xi, \eta) = \xi^T J \eta, \quad J = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$$

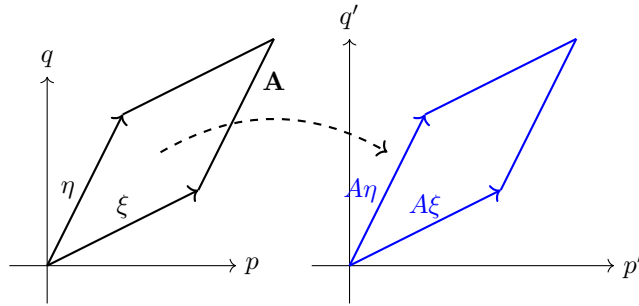
A linear transformation $A : \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ is symplectic if:

$$A^T J A = J$$

Equivalently:

$$\omega(A\xi, A\eta) = \omega(\xi, \eta) \quad \text{for all } \xi, \eta \in \mathbb{R}^{2d}.$$

Interpretation: Symplecticity means that the area of the parallelogram P is preserved by the linear transformation A .



4 Numerical Methods

- Introduction to symplectic integrators
- Benefits in interplanetary dynamics
- Overview of numerical methods
- Stability and computational precision

Since symplecticity is a characteristic feature of Hamiltonian systems, it is natural to seek numerical methods that also possess this property.

A one-step numerical method is called **symplectic** if the mapping for a step $h > 0$:

$$y_1 = \Phi_h(y_0) \quad (3)$$

is symplectic when the method is applied to a smooth Hamiltonian system.

5 Examples of Numerical Methods

- $z_{k+1} = z_k + hf(z_k)$ (explicit Euler)
- $z_{k+1} = z_k + hf(z_{k+1})$ (implicit Euler)
- $z_{k+1} = z_k + hf(q_k, p_{k+1})$ (symplectic Euler)
- $z_{k+1} = z_k + hf\left(\frac{z_{k+1} + z_k}{2}\right)$ (implicit midpoint)
- Runge–Kutta 4th order (not symplectic)
- $I(f) = \int_{-1}^1 f(x)dx \approx \sum_{i=1}^n w_i f(t_i)$

The last one is Gauss–Legendre method, where t_i are the roots of the i th Legendre Polynomial:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n \quad (n = 0, 1, \dots).$$

They can also be expressed in explicit form:

$$P_n(x) = \frac{1}{2^n} \sum_{i=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^i \binom{n}{i} \binom{2n-2i}{n} x^{n-2i}.$$

- **Basic Verlet Algorithm**

$$\begin{aligned} r(t + \Delta t) &= 2r(t) - r(t - \Delta t) + \frac{f(t)}{m} \Delta t^2 \\ v(t) &= \frac{r(t + \Delta t) - r(t - \Delta t)}{2\Delta t} \end{aligned}$$

- **Velocity Verlet Algorithm**

$$r(t + \Delta t) = r(t) + v(t)\Delta t + \frac{f(t)}{2m}\Delta t^2$$

$$v(t + \Delta t) = v(t) + \frac{f(t + \Delta t) + f(t)}{2m}\Delta t$$

Δt time step,

$v(t)$ velocity at time t ,

$r(t)$ position at time t ,

$f(t)$ force at time t .

1. Verlet algorithms are time-reversible.
2. It preserves the symplectic structure of Hamiltonian dynamics (and, for example, the phase flow volume).
3. Generally, it does not conserve the total energy but preserves the value of a perturbed Hamiltonian H' (fluctuations around the prescribed value).
4. It is fast but not highly accurate (classified as an algorithm with global second-order accuracy).
5. It can be fully implemented using only three arrays (positions, velocities, forces).

6 Energy Dissipation in Hamiltonian Systems

It is worth mentioning that in real situations we have energy dissipation.

Energy dissipation is a process in which the mechanical energy of a system is lost—most often converted into thermal energy or other forms that cannot be used to perform work. Such processes occur, for example, due to friction, air resistance, or other non-conservative forces.

Hamiltonian systems are conservative systems in which the total energy, expressed by the Hamiltonian function, remains constant over time, provided that no external non-conservative forces act on the system. This means that in a purely Hamiltonian description, energy dissipation does not occur.

7 Current Progress

Currently I'm working on Python simulation of satellite flying between Earth and Mars. I want to include the influence of gravitational forces of whole Sun system using classical Newtonian mechanics:

$$m_i \frac{d^2 \mathbf{q}_i}{dt^2} = \sum_{j=1, j \neq i}^n \frac{G m_i m_j (\mathbf{q}_j - \mathbf{q}_i)}{\|\mathbf{q}_j - \mathbf{q}_i\|^3}$$

Instead of making not very accurate predictions I will use existing and most reliable data used by NASA. I won't focus on not important details like tidal forces of planet's core, which is beyond the scope of this work, and I will take the planetary and lunar ephemerides DE430 and DE431. These are coordinates generated by fitting numerically integrated orbits of the Moon and planets to observations.

DE431 is a longer integration algorithm (covering years -13,200 to +17,191) than DE430 (covering years 1550 to 2650). The orbits of the inner planets are known to subkilometer accuracy through fitting radio tracking measurements of spacecraft in orbit about them. Very long baseline interferometry measurements of spacecraft at Mars allow the orientation of the ephemeris to be insanely accurate.